
Some F18CD results used in F18CE

Number systems:

Here are the standard abbreviations for the several kind of numbers.

$\mathbb{N} := \{1, 2, 3, \ldots\}$	the set of *natural numbers*
$\mathbb{Z} := \{0, -1, 1, -2, 2, -3, 3, \ldots\}$	the set of *integers*
$\mathbb{Z}^+ := \mathbb{N} = \{1, 2, 3, \ldots\}$	the set of *positive integers*
$\mathbb{Z}^- := -\mathbb{N} = \{-1, -2, -3, \ldots\}$	the set of *negative integers*
$\mathbb{Q} := \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$	the set of *rational numbers*

Properties of the Real Numbers

Definition 3.1 Modulus/Absolute value

The absolute value function $|\cdot|: \mathbb{R} \to \mathbb{R}$ is defined as

$$\left|x\right| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}.$$

Example 3.2.

(i) $|7| = 7$ (ii) $|{-5}| = 5$ (iii) $|0| = 0$.

Absolute value involving inequalities

Case 1: $\left|x-a\right| \leq b \Leftrightarrow -b \leq x-a \leq b \Leftrightarrow a-b \leq x \leq a+b$

Case 2: $\left|x-a\right| \geq b \Leftrightarrow x-a \leq -b$ **or** $x-a \geq b \Leftrightarrow x \leq a-b$ **or** $x \geq a+b$

Remark: The result is true when $\leq$ is replaced by $<$, and $\geq$ is replaced by $>$.

Theorem 3.3: Properties of Absolute Value Function

(i) $|a| \geq 0$, with equality if and only if $a = 0$.

(ii) $|a| = |-a|$.

(iii) $-|a| \leq a \leq |a|$.

(iv) $|a \cdot b| = |a| \cdot |b|$.

(v) $1/|a| = |1/a|$, if $a \neq 0$.

(vi) $|a|/|b| = |a/b|$, if $b \neq 0$.

(vii) $|a| \leq b$ if and only if $-b \leq a \leq b$.

(viii) (Triangle Inequality) $|a + b| \leq |a| + |b|$.

(ix) (Reverse Triangle Inequality) $||a| - |b|| \leq |a - b|$.

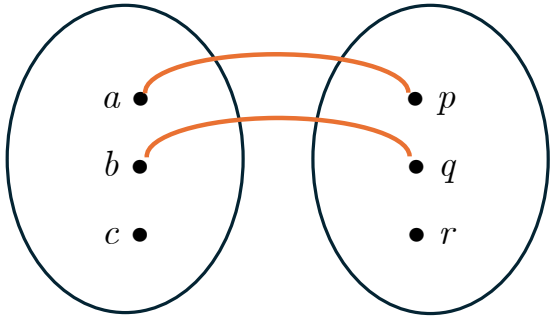
Definition of functions

Let A be a set such that $A \subseteq \mathbb{R}$. A function f from A to $\mathbb{R}$ (often called a 'real-valued' function), written as $f : A \to \mathbb{R}$ is an assignment which map each element $a \in A$ to a unique number $f(a) \in \mathbb{R}$. A is called the domain of f.

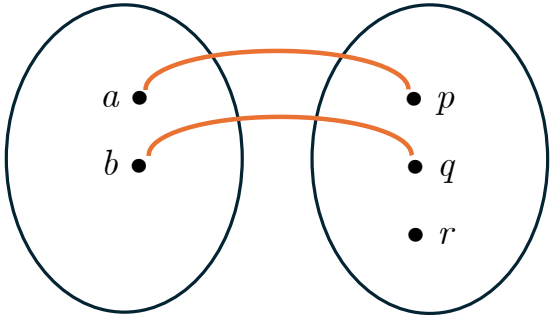
Which one is a function?

Consider a mapping from $\{a, b, c\}$ to $\{p, q, r\}$.

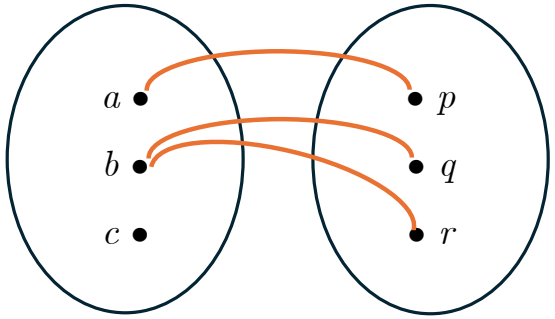
This is not a function.



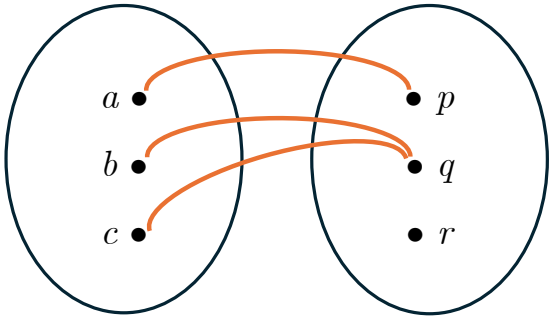
This is a function.



This is not a function.



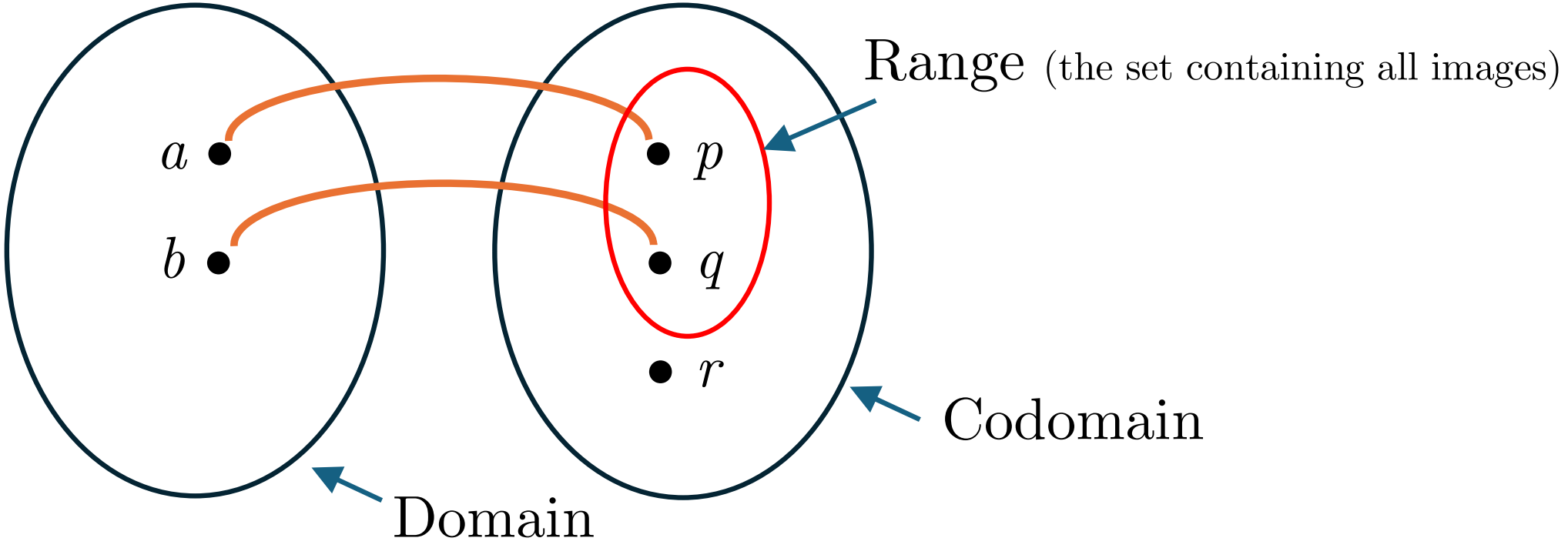
This is a function.



Domain, codomain and range

Let A and B be non-empty sets and let $f : A \to B$ be a function. Then we say domain of f is A and co-domain of f is B. The set A that we define must not be undefined for function f. The range of f is a set $C \subseteq B$, where $C = \{f(a) : \text{for all } a \in A\}$.

Remark: A real function is a function with its codomain is $\mathbb{R}$.


Range (the set containing all images)
a
b
p
q
r
Codomain
Domain

Definition of limit

Let $f : D \to \mathbb{R}$ be a function. Let $c \in \bar{D}$ and $x \in D$.

$$\boxed{\lim_{x \to c} f(x) = L} \text{ if and only if } \boxed{\begin{array}{l} \forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that} \\ 0 < \left|x - c\right| < \delta \Rightarrow \left|f(x) - L\right| < \varepsilon \end{array}}$$

Squeeze theorem in function

Let $f, g, h : D \to \mathbb{R}$ be functions and $c \in \bar{D}$.

Suppose $f(x) \leq g(x) \leq h(x)\ \forall x \in D$ and $\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L$.

Then $\lim_{x \to c} g(x) = L$.

The theorem is also true when changed to "<" (less than).

Property of limits

Let $f, g : D \to \mathbb{R}$ be functions such that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$ for some $a \in \bar{D}$. Let $c \in \mathbb{R}$ be a constant. Then

(a) $\lim_{x \to a} c = c$

(b) $\lim_{x \to a} cf(x) = cL$

(c) $\lim_{x \to a} \Big(f(x) + g(x) \Big) = L + M$

(d) $\lim_{x \to a} \Big(f(x) g(x) \Big) = LM$

(e) $\lim_{x \to a} \Big(f(x) \,/\, g(x) \Big) = L \,/\, M$, provided $g(x) \neq 0$ for all $x \in D$ and $M \neq 0$,

Theorem 7.2

Theorem 7.2. Sequential Criterion vs Function Limits

Let $f : D \to \mathbb{R}$ be a function and $c \in \overline{D}$.

Let $a : \mathbb{N} \to \mathbb{R}$ be a sequence with $a_n \neq c \ \forall n \in \mathbb{N}$. Then

$$\boxed{\lim_{x \to c} f(x) = L} \Leftrightarrow \boxed{\lim\left(f\left(a_n\right)\right) = L \ \forall \left(a_n\right) \text{ with } \lim\left(a_n\right) = c.}$$

Corollary 7.3

When apply the contrapositive, we have:

Let $\left(t_n\right),\left(s_n\right)$ be distinct sequences such that

$$\lim_{n\to\infty}\left(t_n\right)=c=\lim_{n\to\infty}\left(s_n\right),$$

but $\lim_{n\to\infty} f\left(t_n\right)\neq\lim_{n\to\infty} f\left(s_n\right)$ (or limit does not exist).

Then $\lim_{x\to c} f(x)$ does not exist.

Definitions for one-sided limit

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$.

(i) Then f converges to $L \in \mathbb{R}$ as x tends to c *from the right* if

$$\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \ni \left| f(x) - L \right| < \varepsilon \ \text{ whenever } \ c < x < c + \delta \, , \ x \in D \, .}$$

We write this as $\lim\limits_{x \to c^+} f(x) = L$.

(ii) Then f converges to $L \in \mathbb{R}$ as x tends to c *from the left* if

$$\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \ni \left| f(x) - L \right| < \varepsilon \ \text{ whenever } \ c - \delta < x < c \, , \ x \in D \, .}$$

We write this as $\lim\limits_{x \to c^-} f(x) = L$.

Theorem

$$\lim_{x \to a} f(x) \text{ exists and equal to } L$$

if and only if

$$\textit{both } \lim_{x \to a^-} f(x) \text{ and } \lim_{x \to a^+} f(x) \text{ exist and equal to } L$$

Remark: If the left and right limits exist but does not equal, then $\lim_{x \to a} f(x)$ does not exist.

Result

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 .$$

Inequalities results

1. Triangle inequality: $\left|\left|x\right| - \left|y\right|\right| \leq \left|x + y\right| \leq \left|x\right| + \left|y\right| \forall x, y \in \mathbb{R}$.
2. Generalisation: $\left|x_1 + x_2 + \cdots + x_n\right| \leq \left|x_1\right| + \left|x_2\right| + \cdots + \left|x_n\right| \forall x_1, x_2, \ldots, x_n \in \mathbb{R}$.
3. For all $\left|x\right| \leq 1$, $1 \geq \left|x\right| \geq \left|x\right|^2 \geq \cdots \geq \left|x\right|^n \geq \cdots \geq 0$, where $n \in \mathbb{N}$.
6. For all $0 < x, y < 1$, $0 < xy < 1$.
7. For all $x > y > 0$, $0 < \frac{1}{x} < \frac{1}{y}$.
8. $\left|\sin x\right| \leq x$, $\left|\cos x - 1\right| \leq \left|x\right|$, $\left|\cos x\right| \leq 1$, $\left|\sin x\right| \leq 1$, $\left|\tan^{-1} x\right| < \frac{\pi}{2}$ etc.

Function continuity

Let D be an open interval and let $f : D \to \mathbb{R}$ be a function.

Then f is said to be continuous at $x = a \in \mathbb{R}$, if all below are satisfied:

1. $f(a)$ is defined, i.e., $a \in D$.
2. $\lim_{x \to a} f(x)$ exists.
3. $\boxed{\lim_{x \to a} f(x) = f(a)}$, i.e., 1. = 2.

Remark: Some author will only write 3.

Theorem. Properties of continuous functions

Let $f, g : D \to \mathbb{R}$ be continuous functions at the point $c \in D$.

Then the following functions are all continuous at the point c.

(i) $f + g$ (ii) fg (iii) $1/g$, proving $g(c) \neq 0$

(iv) f/g, proving $g(c) \neq 0$ (v) λf , $\lambda \in \mathbb{R}$

Discontinuity

A function is said to be discontinuous at a point $x = a$ if *either one* of the following occur:

1. $f(a)$ is undefined.

2. $\lim_{x \to a} f(x) \nexists$.

3. $f(a)$ is defined and $\lim_{x \to a} f(x) \exists$ but $\lim_{x \to a} f(x) \neq f(a)$.

Definition of Differentiability

Let $f : \Big(a, b\Big) \to \mathbb{R}$ be a function. Then we say that f is differentiable at a point $c \in \Big(a, b\Big)$ if

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$

$f'(c)$ is called as the derivative of f at c.

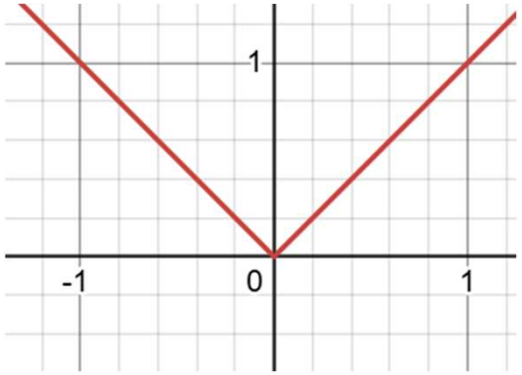
Remark: Alternative definition

Let $x = c + h$. Then f is differentiable at c if

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \text{ exists.}$$

This formula is normally used to find the (first) derivative of a function (the previous one is used to determine the differentiability).

Example



Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = |x|$.

Show that f is not differentiable at 0.

Proof. $\lim\limits_{x\to 0^-} \frac{f(x)-f(0)}{x-0} = \lim\limits_{x\to 0^-} \frac{-x}{x} = -1 \neq 1 = \lim\limits_{x\to 0^+} \frac{x}{x} = \lim\limits_{x\to 0^+} \frac{f(x)-f(0)}{x-0}$,

'sharp' point at 0

so $\lim\limits_{x\to 0} \frac{f(x)-f(0)}{x-0}$ does not exist. Thus, f is not differentiable at 0.

Example

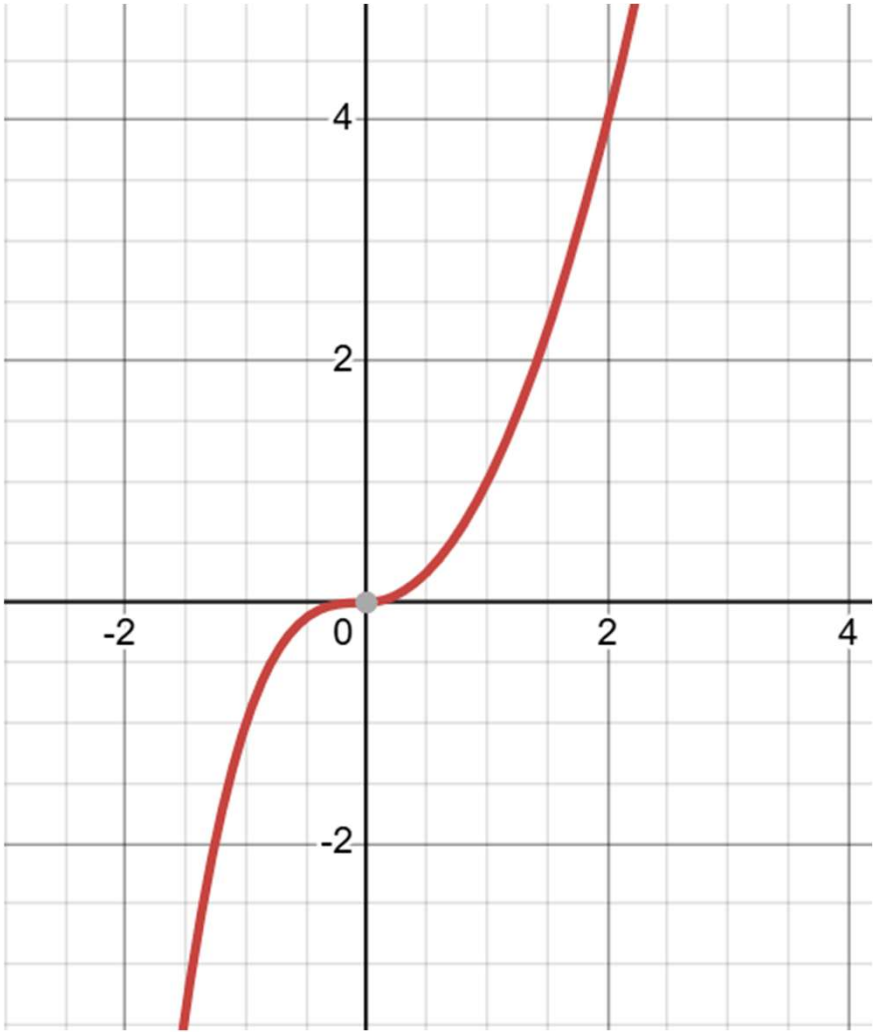
Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \begin{cases} x^3, x < 0 \\ x^2, x \geq 0 \end{cases}$.

Is f differentiable (and continuous) at 0?

Proof. $\lim\limits_{x \to 0^-} \frac{f(x)-f(0)}{x-0} = \lim\limits_{x \to 0^-} \frac{x^3}{x} = 0 = \lim\limits_{x \to 0^+} \frac{x^2}{x} = \lim\limits_{x \to 0^+} \frac{f(x)-f(0)}{x-0}$.

Thus, the limit exists and $f'(0) = 0$.

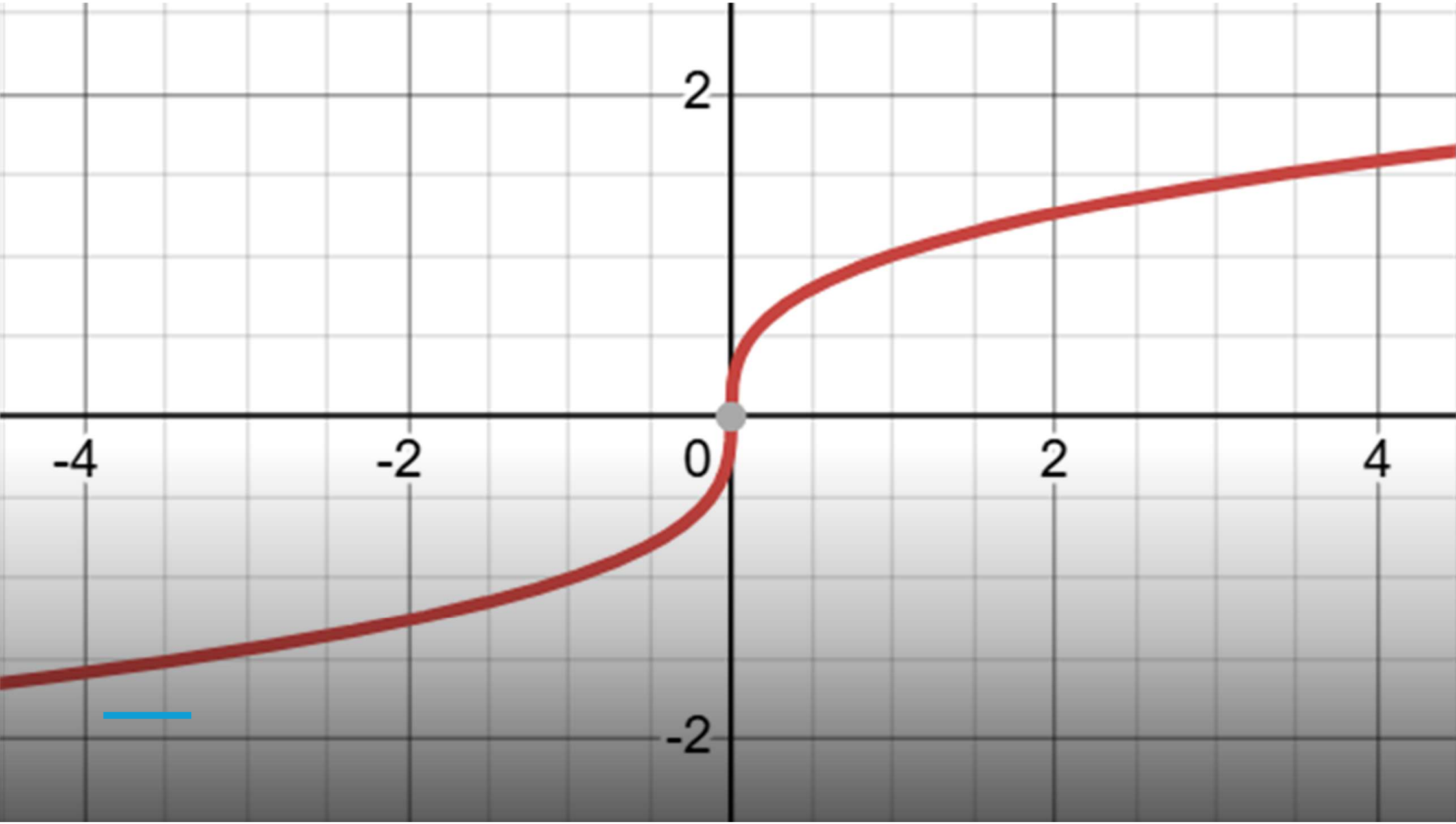
So, f is differentiable (and continuous) at 0.


4
2
-2
0
2
4
-2

Exercise

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = x^{\frac{1}{3}}$.

Is f differentiable at 0?


2
-4
-2
0
2
4
-2

True or False

Geometrically, if a sharp edge can be seen at c, then f is not differentiable at c.

But the converse is false.

Counter example: $f(x) = x^{\frac{1}{3}}$ at 0 is not sharp but it is not differentiable at 0.

Theorem

Let $f : \left(a, b\right) \to \mathbb{R}$ be a function.

If f is differentiable at $c \in \left(a, b\right)$, then f is continuous at c.

Theorem

Let $\lambda \in \mathbb{R}$. Let $f, g : \Big(a, b\Big) \to \mathbb{R}$ be differentiable functions at $c \in \Big(a, b\Big)$.

Then λf, $f + g$, fg, $\frac{f}{g}$ are differentiable functions at c, with derivatives

$$\Big(\lambda f\Big)' = \lambda f' \,(\text{scalar product rule}),\ \Big(f + g\Big)' = f' + g' \,(\text{sum rule}),$$

$$\Big(fg\Big)' = f'g + fg' \,(\text{product rule}),\ \text{and}\ \Big(\frac{f}{g}\Big)' = \frac{f'g - g'f}{g^2}, g \neq 0 \,(\text{quotient rule}).$$

Derivative of polynomial

Any polynomial $p(x) = \sum_{r=0}^{n} a_r x^r$, $a_n \neq 0$ is differentiable on $\mathbb{R}$, with $p'(x) = \sum_{r=1}^{n} r a_r x^{r-1}$.

Theorem 3.7 (The chain rule)

Let $f : D_1 \to \mathbb{R}$ and $g : D_2 \to \mathbb{R}$ with $f(D_1) \subseteq D_2$.

Suppose f is differentiable at $c \in D_1$ and

g is differentiable at $f(c) \in f(D_1)$. Then

$g \circ f : D_1 \to \mathbb{R}$ is differentiable at c with

derivative $\left(g \circ f\right)'(c) = g'(f(c)) \cdot f'(c)$.

True or False

If f is continuous at c, then f is differentiable at c.

False. Counter example: $f(x) = x^{\frac{1}{3}}$ is continuous at 0 but it is not differentiable at 0.

L'Hôpital's rule.

Let $f, g : \left[a, b\right] \to \mathbb{R}$ be continuous functions on $\left[a, b\right]$ and differentiable on $\left(a, b\right)$.

Let $\alpha \in \left[a, b\right]$ such that $f(\alpha) = g(\alpha) = 0$.

Then $\lim\limits_{x \to \alpha} \frac{f'(x)}{g'(x)} = L \Rightarrow \lim\limits_{x \to \alpha} \frac{f(x)}{g(x)} = L$.

Example

Use *L'Hôpital's rule* to find the following limits.

(a) $\lim\limits_{x\to 0}\frac{\ln(x+1)}{x}$ (b) $\lim\limits_{x\to 0}(\tan x \csc x)$

(c) $\lim\limits_{x\to 0^+}\frac{\ln(\ln(x+1)+1)}{x}$ (d) $\lim\limits_{x\to 0}\frac{1-\cos x}{x^2}$

Solution. (a) Since $\ln 1 = 0$, it follows that

$$\lim_{x\to 0}\frac{\ln(x+1)}{x} = \lim_{x\to 0}\frac{1}{x+1} = 1.$$

(b) Since $\tan 0 = 0 = \sin 0$, it follows that

$$\lim_{x\to 0} \frac{\tan x}{\sin x} = \lim_{x\to 0} \frac{\sec^2 x}{\cos x} = \lim_{x\to 0} \frac{1}{\cos^3 x} = 1.$$

(c) Since $\ln(\ln 1 + 1) = 0$, it follows that

$$\lim_{x\to 0^+} \frac{\ln(\ln(x+1)+1)}{x} = \lim_{x\to 0^+} \frac{1/(x+1)}{\ln(x+1)+1} = 1$$

Remark: L'Hôpital's rule can be applied to one sided limit.

(d) Since $1-\cos 0 = 0 = 0^2$, it follows that

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim_{x\to 0} \frac{\sin x}{2x} = \frac{1}{2}\lim_{x\to 0} \frac{\sin x}{x}.$$

Either we can conclude that the above limit is ½ because we already know $\lim_{x\to 0} \frac{\sin x}{x} = 1$ or we may apply L'Hôpital's rule multiple times to obtain the final answer.